

Shower

W21 (2021) — English translation

The material point of mass m is fixed at the end of a weightless rod, which can freely rotate around the horizontal axis O . Initially, the material point is in a position of stable equilibrium under the action of force F , directed perpendicular to the rod (see figure). At some moment, the direction of force F instantly changes to the opposite. Subsequently, the force also remains constant in magnitude and directed perpendicular to the rod, rotating it counterclockwise.

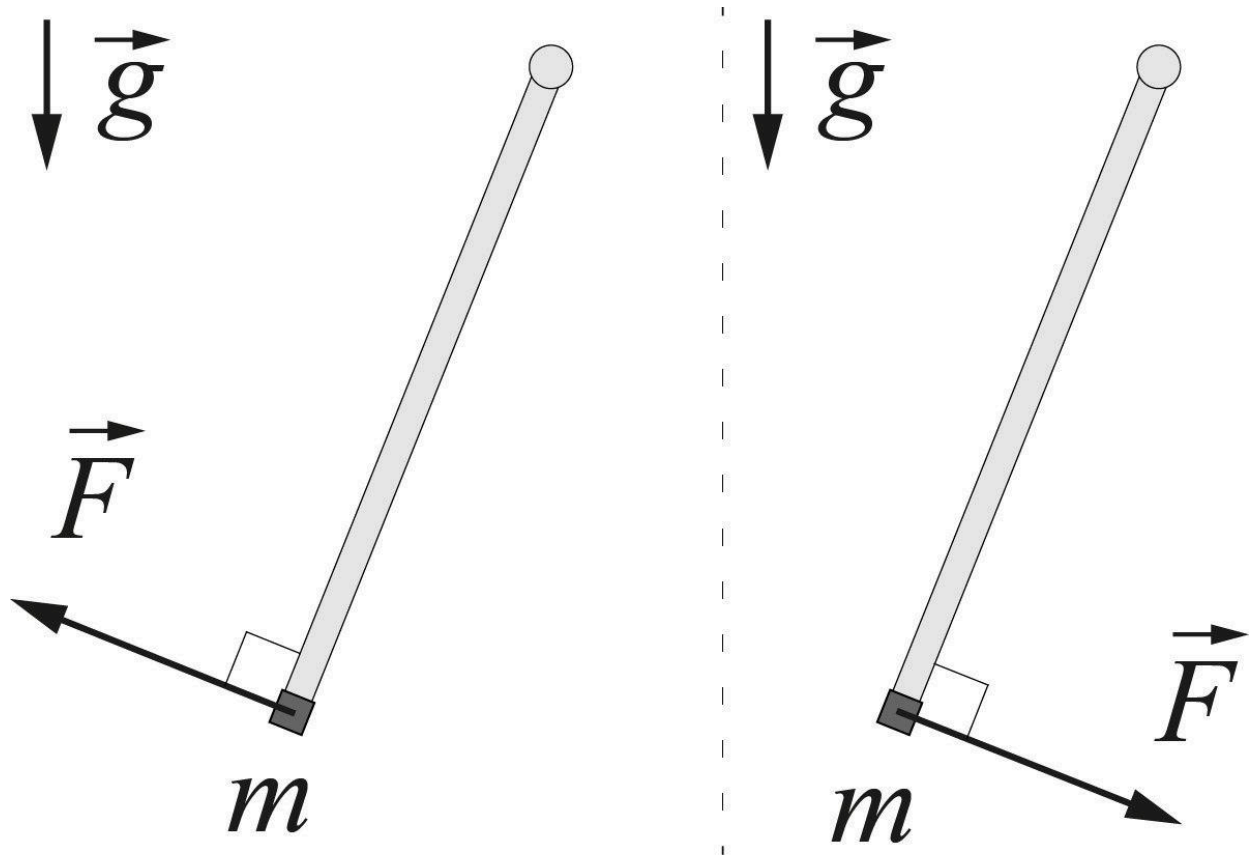


Fig. 1.

A1 At what values of F can the rod make a full revolution around the axis of rotation?

10.00

Source: <https://pho.rs/p/65>